

DiffCoR: Exposing AI-Generated Image by Using Stable Diffusion Model Based on Consistent Representation Learning

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Van-Nhan Tran ; Piljoo Choi ; Hoanh-Su Le ; Suk-Hwan Lee ; Ki-Ryong Kwon All Authors

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Abstract:

Diffusion-based generative models have significantly advanced the field of image synthesis, presenting additional challenges regarding the integrity and authenticity of digital images. Consequently, the identification of AI-generated images has become a critical problem in image forensics. However, there is a lack of literature addressing the detection of images generated by diffusion models. In this article, our focus is on developing a model capable of detecting images generated through both GAN techniques and diffusion models. We propose DiffCoR, a novel detection method for identifying AI-generated images. It consists of two main modules: Stable Diffusion Processing (SDP) and Image Representation Learning (IRL). The SDP module uses a pre-trained Stable Diffusion model to reconstruct input images via reverse diffusion and captures subtle manipulations through reconstruction discrepancies. The IRL module applies self-supervised learning with Latent Consistency Loss (LCL) to extract robust, invariant features, ensuring consistent latent representations across augmented views. We also incorporate frequency domain analysis using Discrete Fourier Transform (DFT) to enhance manipulation detection. Additionally, we introduce ForensicsImage, a publicly available dataset of over 400,000 real and AI-generated images from LSUN-Bedroom, CelebA-HQ, CelebDFv2, and various diffusion models. Experiments on ForensicsImage and GenImage show that DiffCoR achieves state-of-the-art performance, with strong cross-dataset generalization, making it suitable for real-world use in digital forensics, content verification, and social media moderation.

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SECTION I. Introduction

The problem of AI-generated content has gained a lot of attention in recent years. With the recent emergence of several cutting-edge generative models, including Stable Diffusion [1], Imagen [2], DALL-E 2 [3], Denoising Diffusion Implicit Models (DDIM) [4], Denoising Diffusion Probabilistic Models (DDPM) [5], and variety of diffusion models (DM) [6], [7] have gained popularity for the creation of digital content. One of the most significant challenges in the development of general and human-like artificial intelligence is creating an automated system that can create visually realistic images from textual descriptions [8]. However, the misuse of image-generation tools facilitates the creation of fake images, leading to potential public deception. This underscores the pressing need to safeguard the integrity and authenticity of images [9]. Furthermore, given the rapid progress in face manipulation techniques, including autoencoder-based [10] and GAN-based [11], [12] methods, synthetic faces now closely resemble real faces, making it difficult for humans to distinguish between them. This presents a substantial risk to societal trust and security. Therefore, it is important to create efficient methods for detecting fake content. In recent years, many proposed methods have focused on Deepfake detection. Many works can be mentioned as Deepfake detection through biological



evidence [13], offered structures that applied learning-based methods in [14], [15] extract features from the spatial domain and have demonstrated exceptional performance on particular datasets; Face forgery detection based on the frequency domain [16], [17] provide state-of-the-art performance by using local frequency statistics and frequency-aware decomposition. A few studies that are worthy of notice include diffusion-generated image detection [18], [19], and the detection of fake images generated by the text-to-image technique [20], [21]. Techniques capable of detecting both deepfake images and those generated by diffusion models are still relatively new. This area is garnering significant attention from the research community.

In this article, we propose DiffCoR (Stable Diffusion Model Based on Consistent Representation Learning), a framework for detecting AI-generated images, including those produced by diffusion models and Deepfakes. DiffCoR consists of two main components: Stable Diffusion Processing (SDP) and Image Representation Learning (IRL). The SDP module uses an encoder and a pre-trained Stable Diffusion model to reconstruct input images from their noisy versions, then computes discrepancies between the input and reconstructed images to detect subtle manipulations, an approach inspired by DIRE [18], but without relying on model-specific tuning. The IRL module learns robust features for classification using a deep encoder and applies Latent Consistency Loss (LCL) to ensure consistent representations under different input transformations. The key contributions of this work are summarized as follows:

- We propose DiffCoR, a novel detection method for AI-generated images. The DiffCoR specifically takes advantage of the unique properties of diffusion models and utilizes the method of representation learning to classify manipulated images.
- We introduce Latent Consistency Loss (LCL), designed to ensure that the latent representations of similar inputs are proximate in the latent space, thereby promoting improved structure and coherence in the learned representations.
- We perform conversion and analysis of both original and AI-generated images in the frequency domain utilizing the Discrete Fourier Transform (DFT). This approach facilitates a deeper understanding of AI-generated imagery and improves the accuracy of detecting manipulated images.
- We created a new dataset named ForensicsImage to benchmark AI-generated image detectors.
- We evaluate the proposed DiffCoR with extensive experiments on the ForensicsImage dataset and compare it to many recent approaches. The results prove that the proposed DiffCoR achieves state-of-the-art performance in detecting AI-generated images.

SECTION II. Related Work

A. Generative Diffusion Models for Image Generation

Generative Diffusion Models have gained significant attention in machine learning for their ability to generate high-quality data by progressively denoising random noise. Unlike GANs [23], diffusion models [22], first introduced by Sohl-Dickstein et al., offer stable training and strong theoretical grounding. When applying diffusion models to image generation, three primary categories must be considered: 1)

Unconditional & class condition. Key variants include DDPM [5], known for robust and high-quality image generation; DDIM [4], which improves sampling efficiency and flexibility; and ADM [6], which enhances unconditional image synthesis using classifier guidance for better sample quality. 2) **Text condition.** Stable Diffusion [1] introduced a method that involves the controlled diffusion of noise over multiple steps to generate realistic images. This approach contrasts with traditional generative models like GANs or VAEs, which directly generate images from random noise. DALL-E [3] is specifically tailored for generating images from textual descriptions. It builds upon the transformer architecture. Imagen [2] presented a novel text-to-image diffusion model characterized by a superior level of linguistic comprehension and exceptional photorealistic rendering capabilities. Imagen excels in generating images of high fidelity, further enhanced by the advanced text comprehension abilities ingrained in large-scale transformer language models. The work of HERTZ, Amir, et al. [24] introduced an innovative prompt-to-prompt editing framework that enables text-based control over image editing. Furthermore, many studies in this field can be mentioned, Dreambooth [25], Attend-and-excite [26], etc. 3) **Image Condition.** Instructpix2pix [27] introduced an approach for image editing based on human-written instructions, leveraging pre-existing models, Stable Diffusion and GPT-3. ControlNet [28] utilizes existing diffusion models to gradually expand parameters without introducing harmful noise during fine-tuning. ControlNet enables diverse conditional controls, such as depth, edges, segmentation, and human pose, to be applied to diffusion models.

B. AI-Generated Image Detection

In the realm of computer vision, the identification of synthesized images has been a prominent area of research. Prior methodologies have concentrated on discerning specific types of manipulations, such as copy-move [29], splicing [30], and inpainting [31]. Subsequently, deep learning methodologies have been employed to discern sophisticated forgeries generated by diffusion models or Generative Adversarial Networks (GANs) [32]. Some research focused on Deepfake detection. It involves the development of techniques and tools to identify synthetic media created using deep learning algorithms. Face X-ray [33] presented a method for fake face image detection. This representation is a grayscale image that indicates whether the input image is a composite of two images from different sources. ICT [34] introduced a method for Deepfake detection, which applied a consistency loss and a Transformer architecture. Multi-attention [35] presented a multi-attentional fake face detection. Fusing [36] introduced a dual-branch model designed to detect fake images produced by various GAN techniques. Similarly, GramNet [37] proposed an architecture that leverages global image texture representation to identify fake images generated by a range of GAN models. For identifying images produced by GANs in the frequency domain, CNN-DCT [38] presented a method to investigate the fake image in frequency and use it to classify these images.

C. Consistency Learning

Consistency learning is a technique used particularly in the context of semi-supervised learning and unsupervised learning. The primary goal of consistency learning is to ensure that the predictions of the model remain stable or consistent under various perturbations or transformations of input data. Some popular methods can be indicated. Pi-Model [39], a straightforward consistency learning strategy wherein the model is trained to produce outputs that are consistent for both the original and augmented data. VAT [40] uses adversarial perturbations to enforce consistency, improving robustness to small input changes. CatGAN [41] introduced a method that teaches a discriminative classifier how to learn from data that is either partially or not labeled. We apply consistency learning to manipulated image detection and demonstrate that it is also beneficial for AI-generated image detection.

D. Contrastive Learning

Contrastive learning is widely used in the context of self-supervised learning, where labeled data is scarce or unavailable. This approach is used primarily for representation learning, where the goal is to learn useful features from data by contrasting positive and negative samples. SimCLR [42] uses data augmentations to generate positive pairs and maximizes the agreement between these pairs. BYOL [43], this method avoids the use of negative samples by employing two neural networks. MoCo [44], maintains a dynamic dictionary with a queue and a moving-average encoder, allowing the use of a large number of negative samples. SwAV [45], this approach ensures consistency across cluster assignments derived from various views of the same image while clustering the data concurrently.

SECTION III. Method

This section presents the proposed DiffCoR framework for AI-generated image detection, which consists of two key components: Stable Diffusion Processing (SDP) and Image Representation Learning (IRL) that work together to improve the detection of AI-generated images. The SDP module computes the difference between the input image x_0 and the reconstructed output x_1 using a pre-trained Stable Diffusion v2 (SDv2) model [1]. It reconstructs the input image via a reverse diffusion process and computes the difference between the original and reconstructed outputs. This difference image captures subtle semantic and frequency-domain inconsistencies introduced by generative models, serving as a residual representation that highlights manipulation artifacts. The IRL module, inspired by BYOL [43], learns robust representations by comparing the original image I_0 with its augmented version I_1 . The IRL module then takes this difference as input and learns robust, invariant feature representations using a self-supervised strategy, guided by the Latent Consistency Loss (LCL). Through this interaction, SDP provides meaningful manipulation-sensitive inputs, while IRL ensures that the extracted features are both discriminative and generalizable across unseen generative domains. This complementary design enhances the ability of the model to capture high-level artifacts that may not be evident in raw pixel space alone. This combination enables DiffCoR to generalize effectively to images generated by diverse or previously unseen AI techniques. An overview of the framework is illustrated in Fig. 2. The remainder of this section covers: (1) the diffusion model process in SDv2, (2) the SDP module, (3) the IRL module, (4) the proposed dataset, and (5) the loss function design.

A. Preliminaries

In this section, we explain the background of the diffusion model. In a diffusion model [22], there are two main components: an encoder and a decoder. The diffusion model is shown in Fig. 1. The encoder receives a

data sample x and passes it through a sequence of intermediate latent variables $z_1, z_2, \dots, z_{T-1}, z_T$. Clean input data are gradually corrupted by noise over multiple steps according to a schedule $\beta_1, \dots, \beta_T$. After a sufficient number of steps, the conditional distribution $q(z_T|x)$ and the marginal distribution $q(z_T)$ of the final latent variable converges to the standard normal distribution. As this process is predetermined, all the learned parameters are located within the decoder. The joint distribution of all of the latent variables $z_1, z_2, \dots, z_T$ given input x is represented as:

$$q(z_{1..T}|x) = q(z_1|x) \prod_{t=2}^T q(z_t|z_{t-1}),$$

$$q(z_t|z_{t-1}) := N(z_t; \sqrt{1 - \beta_t} \cdot x_{t-1}, \beta_t I) \quad (1)$$

► View Source ?

where $N(\cdot)$ is the operation of normal distribution, the identity matrix, denoted as I .

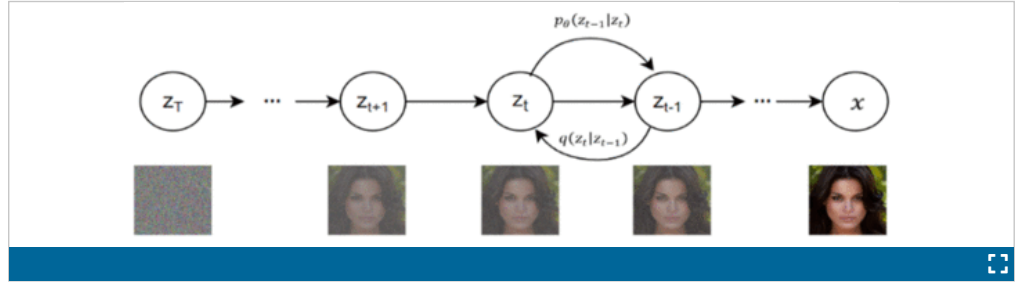


Figure 1.
An example of a diffusion model.

By learning to reverse this diffusion process, the decoder performs the reverse operation, starting with z_T and progresses backward through $z_{T-1}, \dots, z_1$ until it ultimately reconstructs the original data point x . Both the encoder and decoder operate stochastically, meaning that their mappings involve randomness rather than strict determinism. The reverse process is represented as:

$$p_\theta(x \dots z_T) := p_\theta(z_T) \prod_{t=2}^T p_\theta(z_{t-1}|z_t) \quad (2)$$

► View Source ?

where $p_\theta(x \dots z_T)$ is the joint distribution, the starting point of the learned Gaussian transition is $p(z_T) = N(z_T, 0, I)$.

Stable Diffusion Model (SDv2) [1]. SDv2 improves upon traditional diffusion models by addressing training instability through enhanced noise scheduling and network design. These improvements lead to more reliable training, reduce the risk of mode collapse, and enable faster convergence. As a result, SDv2 consistently produces high-quality samples with greater efficiency.

B. Stable Diffusion Processing

This section is based on the architecture of Stable Diffusion [1]. The SDP utilizes a pre-trained Stable Diffusion v2 model to process the input image via an encoder and a diffusion process that adds noise in a controlled way. This module then reconstructs the image using DDIM inversion and computes the difference between the original and reconstructed outputs. This discrepancy is used to identify manipulations in the image that may not be visually obvious but are exposed during the inversion-reconstruction cycle. Unlike previous approaches that rely on model-specific reconstruction paths, our method leverages the consistent behavior of Stable Diffusion, improving robustness across unknown generative sources. The architecture of the SDP module is shown in Fig. 2. The encoder E converts an input image $x_0 \in \mathbb{R}^{H \times W \times 3}$ into the latent representation $z_0 = E(x_0)$. The mechanism of the encoder E is based on the architecture of an autoencoder, described in previous work [46]. A CNN encoder is used to represent images in code. We use the DDIM inverse scheduler [4] in the Diffusion Process module. It progressively combines the input with white noise samples. The latent representation z_0 is mapped through a series of intermediate samples $z_1, z_2, \dots, z_T$. The z_T is presented as:

$$z_T = \sqrt{1 - \beta_{T-1}} \cdot z_{T-1} + \sqrt{\beta_T} \cdot G_T \quad (3)$$

[View Source](#)

where G_T is noise sampled from a standard normal distribution. The first term reduces the data magnitude and accumulated noise, while the second adds new noise, controlled by the hyperparameter $\beta_T \in [0, 1]$.

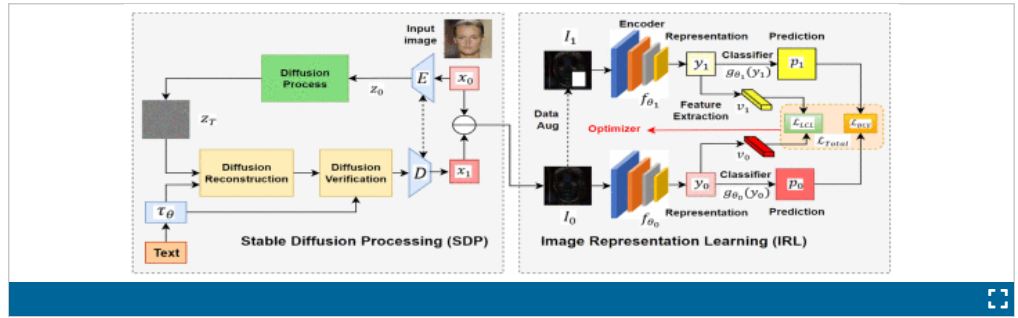


Figure 2.
Overall architecture of our proposed DiffCoR.

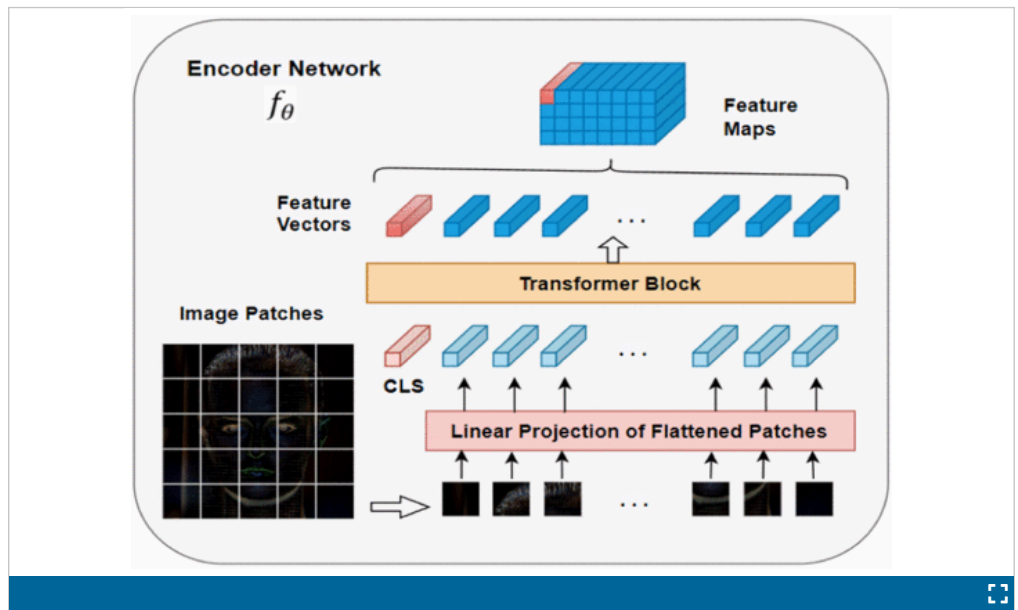


Figure 3.
Illustration of feature extraction for images using the ViT model.

The latent representation z_T is passed through Diffusion Reconstruction (DR) and Diffusion Verification (DV) modules using a pre-trained Stable Diffusion v2 (SDv2) model [1]. A text prompt is included as a condition, but we leave it empty to treat this as an image reconstruction task. The decoder D then reconstructs the image as $x_1 = D(\hat{x}_0)$. The final output is the pixel-wise difference between the original x_0 and reconstructed x_1 , where m, n represent corresponding pixel positions. The output of the SDP module is formally defined as:

$$I_0(m, n) = |x_1(m, n) - x_0(m, n)| = |D(\hat{x}_0)(m, n) - x_0(m, n)| \quad (4)$$

[View Source](#)

We apply the Discrete Fourier Transform (DFT) [4,7] to decompose the image $I_0 \in \mathbb{R}^{H \times W \times 3}$ (The width and height of the image are denoted by W and H , respectively). The image I_0 contains three channels. To generate its frequency representation, we first do the following 2D DFT on each channel. The formula for each channel is as follows:

$$\mathcal{F}(u, v) = \sum_{m=0}^{W-1} \sum_{n=0}^{H-1} I(m, n) \cdot e^{-i2\pi \left(\frac{um}{W} + \frac{vn}{H} \right)} \quad (5)$$

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Then, the spectrum for each channel is represented as follows:

$$|\mathcal{F}(u, v)| = \text{mod}(\mathcal{F}(u, v)) \quad (6)$$

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where $\text{mod}(\cdot)$ represents the modulus (magnitude) of a complex number.

We consider the convolutional operation of a 2D generative model. According to the study of the spatial frequency of the generative model [48], the convolutional layer Conv_i^l with the l^{th} layer and output channels i can be represented as:

$$H_i^{l+1} = \text{Conv}_i^l(H^l) = \sigma \left(\sum_{c=1}^{C_{(l)}} F_{ic}^l \otimes \text{Up}(H_c^l) \right) \quad (7)$$

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where $\sigma(\cdot)$ denotes the activation function, $C_{(l)}$ is the number of input channels, F^l is the parameters of the 2D filter, $\otimes$ is a convolutional operator, and $\text{Up}(\cdot)$ denotes the up-sampling operator. If we confine $\sigma(\cdot)$ to Rectified Linear Units (ReLU) in the positive domain, we consider the up-sampling $\text{Up}(\sigma(\cdot))$ as a linear operation. Eq. (7) simply becomes:

$$H_i^{l+1} = \text{Conv}_i^l(H^l) = \sum_{c=1}^{C_{(l)}} F_{ic}^l \otimes H_c^l \quad (8)$$

[View Source](#) 

Upon applying the DFT to H_i^{l+1} . It is now observed in the frequency domain as follows:

$$\begin{aligned} \mathcal{F}(H_i^{l+1}) &= \mathcal{F}(\text{Conv}_i^l(H^l)) = \mathcal{F} \left(\sum_{c=1}^{C_{(l)}} F_{ic}^l \otimes H_c^l \right) \\ &= \sum_{c=1}^{C_{(l)}} \mathcal{F}(F_{ic}^l \otimes H_c^l) = \sum_{c=1}^{C_{(l)}} \mathcal{F}(F_{ic}^l) \times \mathcal{F}(H_c^l) \end{aligned} \quad (9)$$

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We calculate the spectrum difference of the red and green channels. The same applies to the other two cases: green-blue channels and blue-red channels. The difference between red and green channels is represented as follows:

$$\Delta \mathcal{F}_{RG} = \frac{1}{H \cdot W} \sum_{u=1}^W \sum_{v=1}^H ||\mathcal{F}(u, v)|_R - |\mathcal{F}(u, v)|_G| \quad (10)$$

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Then, we calculate the mean of these differences:

$$\text{Mean} = \frac{\Delta \mathcal{F}_{RG} + \Delta \mathcal{F}_{GB} + \Delta \mathcal{F}_{BR}}{3} \quad (11)$$

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The visual representation of our end-to-end pipeline for extracting the aforementioned frequency descriptive features from a given image is depicted in Figs. 5 and 6. These sections present a visual analysis conducted in the frequency domain. The DFT is integrated into the SDP module to capture subtle spectral discrepancies between real and AI-generated images. These discrepancies are often imperceptible in the spatial domain but become more pronounced in the frequency domain, particularly in the distribution of low-frequency and mid-frequency components, which tend to differ systematically between original and manipulated images. High-frequency components, which often contain less discriminative information and may introduce noise, are removed prior to reconstruction in the spatial domain.

C. Image Representation Learning

The IRL is designed to learn robust and invariant representations through self-supervised learning. It applies augmentations to the input and learns consistent feature representations through a dual-encoder structure. To enhance this process, we introduce the Latent Consistency Loss (LCL), which ensures that representations of augmented views remain closely aligned in the latent space, facilitating ability of the model to generalize across both known and novel generative manipulations. Given an input image I_0 with a batch of N samples, we first apply data augmentations τ to obtain N samples of images $I_1 = \tau(I_0)$. These images are fed into encoder networks f_{θ_0} and f_{θ_1} , which is defined by a set of parameters θ_0 and θ_1 . Both networks share the same architecture. These encoders generate two feature vectors $v_0 = f_{\theta_0}^{pre}(I_0)$ and $v_1 = f_{\theta_1}^{pre}(I_1)$. Where f^{pre} represents the network without the last layers and f represents the full network. Therefore, the representation vectors can be written as: $y_0 = f_{\theta_0}(I_0)$ and $y_1 = f_{\theta_1}(I_1)$. The representation vectors are put into the classifier g_{θ_0} and g_{θ_1} for classification. The feature maps are used to calculate latent consistency loss. The classifier network, denoted as g_{θ_0} and g_{θ_1} , consists of a linear layer followed by a softmax normalization layer. This network predicts a representation vector y_0 and y_1 to a predicted probability $p_0 = g_{\theta_0}(y_0)$ and $p_1 = g_{\theta_1}(y_1)$. The resulting probability is used for classification computation.

The structure of the ViT model used in this proposal is shown in Figure. The input image $I_0 \in \mathbb{R}^{H \times W \times 3}$ is divided into a sequence of patches. These patches are used in the linear projection of flattened patch blocks. The CLS token is also created, which is similar to the position embeddings used in the study of ViT [49]. The Transformer Block applies the encoder of the transformer to generate a list of feature vectors. These feature vectors are collected to create feature maps.

D. Loss Functions

Latent Consistency Loss (LCL): The methods of contractive learning often minimize the representation distance across different augmentations of data I_0 and I_1 without considering semantic information. Our method takes a different approach; our method minimizes the representation distance across local regions, but only if it corresponds to the same latent space. The feature maps are collected from the encoder networks f_{θ} . Assume we apply the ViT model. The dimensions of the obtained features are 4D tensors $v_0, v_1 \in \mathbb{R}^{m \times n \times p \times q}$. We perform element-wise multiplication of corresponding elements of tensors v_0 and v_1 . After the element-wise multiplication, we sum the elements of the resulting tensor along the axes n , p , and q , effectively collapsing those axes. This operation results in a new tensor $c \in \mathbb{R}^m$. The equation of the results $c = \{c_i, i \in m\}$ is as:

$$c_i = \sum_{j=1}^n \sum_{k=1}^p \sum_{l=1}^q v_{ijkl}^0 \times v_{ijkl}^1 \quad 1 \leq i \leq m \quad (12)$$

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The LCL aims to encourage the model to produce consistent outputs under different transformations of the input data. The LCL is applied to penalize the distance between two latent vectors v_0 and v_1 . For N pairs of input images, the LCL can be defined as:

$$\mathcal{L}_{LCL} = \sum_{n=1}^N \left(1 - \frac{c_n}{\|c_n\|_2} \right) \quad (13)$$

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The Latent Consistency Loss (LCL) functions similarly to cosine similarity, encouraging aligned but not identical feature representations across augmented views. Unlike L1 or L2 losses, which penalize magnitude differences and enforce identical vectors, LCL focuses on directional similarity, making it more suitable for handling variations such as occlusions (as shown in Fig. 2). This design allows the model to focus on semantically meaningful features while being robust to distortions. LCL is particularly effective in high-

dimensional spaces like those produced by Vision Transformers (ViT), guiding the network toward more generalizable representations.

Classification Loss: We use Binary Cross Entropy loss (BCE) for the binary classification task to measure the difference between the predicted probabilities and the actual binary labels. For N pairs of input images, y_i is the actual label for the i^{th} sample, p_i is the predicted probability for the i^{th} sample. The loss is given by:

$$\mathcal{L}_{BCE} = - \sum_{i=1}^{2N} [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] \quad (14)$$

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Total Loss: We integrate the latent consistency loss with the classification loss to formulate the total loss:

$$\mathcal{L}_{Total} = \mathcal{L}_{BCE} + \lambda \mathcal{L}_{LCL} \quad (15)$$

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where λ is a parameter to balance between the LCL and the BCE.

E. ForensicsImage: An AI-Generated Image Dataset

To more effectively evaluate the performance of the DiffCoR, we have created a dataset, named ForensicsImage. This dataset consists of images generated by various diffusion models and a Deepfake dataset for extensive experimentation. The composition of ForensicsImage is detailed in Table 1. The ForensicsImage dataset is composed of a total of 439901 images, including both real and AI-generated content across diverse domains. Specifically, the dataset contains 117438 real images collected from publicly available datasets such as LSUN-Bedroom [50], CelebA-HQ [51], and unaltered frames from CelebDFv2 [10]. These represent a mix of scene and facial images to ensure visual diversity. The 322463 fake images were generated using a wide range of state-of-the-art generative models, including StyleGAN2, BigGAN, ADM, DDPM, GLIDE, LDM-t2i, Midjourney, DALL-E, and Imagen. These models span various generation paradigms, including unconditional generation and text-to-image synthesis, allowing the dataset to simulate real-world variability in generative techniques. All images are preprocessed to a standard resolution of 256×256 pixels via center cropping and resizing, ensuring uniformity during training and evaluation.

TABLE 1 The Details of the ForensicsImage and GenImage Datasets

Dataset	Condition	Method	Train set (# of images)	Test set (# of images)
LSUN- Bedroom		real	40,000	1,000
	Unconditional	LDM [1]	40,000	1,000
		DDIM [4]	40,000	1,000
		DDPM [5]	-	1,000
		IDDPM [7]	-	1,000
		GD [6]	-	1,000
	Text-to-image	LDM-t2i [1]	40,000	1,000
CelebA- HQ		real	28,000	1,000
	Unconditional	LDM [1]	29,800	1,000
		DALL-E [3]	-	1,000
		Imagen [2]	-	1,000
	Text-to-image	Midjourney [52]	-	1,000
Celeb- DFv2		real [10]	9,396	2,323
		fake [10]	66,432	4,256
AI-Gen		LSUN real	10,000	1,000
		CelebA-HQ real	10,000	1,000
		CelebDFv2 real	9,396	2,323
	Unconditional	LSUN LDM	10,000	1,000
		LSUN DDIM	10,000	1,000
		CelebA-HQ LDM	10,000	1,000
		CelebDFv2 fake	10,000	4,256
	Text-to-image	LSUN LDM-t2i	10,000	1,000
GenImage [53]		real	1,331,167	
		fake	1,350,000	

The proposed dataset comprises unconditional and text-to-image produced images.

TABLE 2 Details Regarding the Data Augmentation Configuration

Name	Data augmentation techniques
NoAug	Resize Normalize
RRCAug	Random Resized Crop Random Horizontal Flip Random Noise Normalize
REAug	Resize Random Horizontal Flip Random Erasing Gaussian blur Normalize
RanSelect	Randomly select from NoAug, RRCAug, and REAug.

From an ethical standpoint, all real images were sourced from open-access datasets intended for academic research, and all fake images were generated in-house using publicly released models. The dataset does not contain any private, sensitive, or personally identifiable information and is shared solely for non-commercial academic research purposes in the domain of digital image forensics. These measures ensure that ForensicsImage adheres to responsible data handling and supports reproducible and ethical AI research.

LSUN-Bedroom: The LSUN-Bedroom is a large-scale dataset widely used in computer vision research, particularly for scene understanding, image classification, and generative model training. The real images are collected from the original LSUN dataset [50], specifically from the bedroom category. We gather images generated by 6 diffusion models. Among these, five subsets (LDM [1], DDIM [4], DDPM [5], IDDPDM [7], GD [6]) are generated by unconditional diffusion models, while the remaining are generated by text-to-image diffusion models (LDM-t2i [1]). The prompt applied for text-to-image generation is “A bedroom”. The quantities of real and generated images within each part are detailed in Table 1.

CelebA-HQ: The CelebA-HQ [51] is a high-quality version of the CelebA dataset, which is widely used in computer vision and machine learning research, particularly for tasks involving facial recognition, image synthesis, and generative models. We collect real images from the CelebA-HQ. We further gather images produced by a variety of methods, LDM [1], DALL-E [3], Imagen [2], and Midjourney [52].

CelebDFv2: The CelebDFv2 is designed to provide a challenging dataset for evaluating and developing deepfake detection algorithms. We utilized videos proposed by the authors [10], the number of videos in both the training and test sets remains unchanged from the original settings. Based on the videos, we extract the faces using the MTCNN [54] library and resizing them to 256×256 pixels.

AI-Gen: We gather data from three datasets above, LSUN-Bedroom, CelebA-HQ, and CelebDFv2. Besides evaluating each separate dataset, we design this dataset to evaluate the generalization of the model. For in-dataset evaluation, we conduct training and testing experiments using real images (LSUN real [50], CelebA-HQ real [51], and CelebDFv2 real [10]) and manipulated images generated by several generative models (LSUN LDM [1], LSUN DDIM [4], CelebA-HQ LDM [1], CelebDFv2 fake [10], and LSUN LDM-t2i [1]). For cross-dataset evaluation, we perform experiments on LSUN DDPM [5], LSUN IDDPDM [7], and LSUN GD [6].

GenImage: [53]. This dataset is created by eight generative models, including: ADM [6], BigGAN [55], GLIDE [56], Midjourney [52], Stable Diffusion v1.4 [1], Stable Diffusion v1.5 [1], VQDM [57], Wukong [58]. This dataset consists of more than one million real and generated image pairs. We use the same settings for training and testing as in the original paper.

SECTION IV. Experiments

A. Implementation Details

Experimental settings: All experiments are conducted using our ForensicsImage and the GenImage [53] datasets. In the SDP, we utilize the pre-trained SDv2 model, employing the large variant for images equal to or larger than 512×512 , and the nano variant for images larger than 256×256 . In the Diffusion Process, Diffusion Reconstruction, and Diffusion Verification modules, we employ a total of 250 diffusion steps by default.

Data augmentation: We use two basic transformations for data augmentation: Random Erasing and Random Resized Crop, along with two more complex augmentation techniques. We use an aspect ratio of (0.4, 2.5) and a scale factor of (0.03, 0.3) for the Random Erasing. We use an aspect ratio of (0.8, 1.0) and a scale factor of (1/1.2, 1.1) for the Random Resized Crop transformation. A 1/3 chance of no augmentation, a 1/3 chance of applying Random Erasing, and a 1/3 chance of applying Random Resized Crop are used for each input image. This augmentation strategy is denoted as RanSelect. Moreover, we apply an additional sophisticated data augmentation method that combines random shift, Gaussian blur, random scale, and random noise. Refer to [59] for additional details.

Evaluation Metrics: We primarily report the Area Under Curve (AUC) as our main evaluation metric. Additionally, we provide accuracy (ACC) as a secondary metric to offer a comprehensive assessment of model performance. AUC evaluates the ability of the model to discriminate between positive and negative instances across various thresholds.

Baselines: 1) EfficientNetBo [60] is a state-of-the-art convolutional neural network architecture optimized for high performance with relatively low computational costs. we use this model as a baseline model for evaluation. 2) GramNet [37] is a method that is proposed for the fake face detection problem. 3) Fusing [36] designed a two-branch model with an attention-based patch selection for AI-generated image detection. 4) DIRE [18] developed a detector that can discriminate between actual and diffusion-generated images. 5) CNN-DCT [38] proposed a method for detecting GAN-generated images by converting them to a frequency domain.

B. Intra-Dataset Evaluation

In this section, we assess the effectiveness of the DiffCoR on the proposed ForensicsImage dataset. We compare our proposal with recent approaches, including EfficientNetBo [60], GramNet [37], Fusing [36], DIRE [18], and CNN-DCT [38]. These techniques show strong results in identifying fake images generated by various GANs and diffusion models. For intra-dataset evaluation, we conduct experiments using the same training and testing sets of the proposed dataset.

The LSUN-Bedroom dataset from the ForensicsImage is used for our initial evaluation. Table 3 indicates the quantitative results. The DiffCoR achieves superior performance across all generative methods, attaining perfect or near-perfect scores. It consistently scores 100.0 in both AUC and ACC for the LDM, and close to perfect scores in the DDIM (AUC 99.8, ACC 99.7) and the LDM-t2i (AUC 99.7, ACC 99.3). These Fusing [36], DIRE [18], and CNN-DCT [38] all achieve high results, indicating that the dataset is not sufficiently challenging and these methods are well-designed to address this issue effectively. The suboptimal results of the GramNet [37] can be attributed to the method being specifically designed for fake face detection.

TABLE 3 In-Dataset Evaluation on the LSUN-Bedroom Subset of the ForensicsImage Dataset

Generative Method Method	LDM		DDIM		LDM-t2i	
	AUC	ACC	AUC	ACC	AUC	ACC
EfficientNetBo [60]	90.1	89.2	92.3	90.5	87.3	83.1
GramNet [37]	87.1	83.8	89.3	85.3	85.6	83.1
Fusing [36]	94.7	93.3	93.4	92.5	93.9	91.2
DIRE [18]	99.1	98.8	100.0	100.0	96.8	95.4
CNN-DCT [38]	97.5	96.6	97.7	96.8	95.3	93.5
DiffCoR (Ours)	100.0	100.0	99.8	99.7	99.7	99.3

Comparison of our DiffCoR model and other image detectors.

For detecting manipulated face images, we run experiments on the CelebA-HQ and CelebDFv2 datasets. The results are shown in Tables 4 and 5. The DiffCoR performs exceptionally well in identifying modified images within the CelebA-HQ, achieving perfect scores with an AUC of 100.0 and an ACC of 100.0. Furthermore, the DiffCoR achieves the highest AUC of 88.9 in the CelebDFv2, indicating strong performance in detecting manipulated faces. The EfficientNetBo [60] shows a lower performance among the models compared, although it still achieves relatively high scores. The GramNet [37], Fusing [37], and DIRE [18] models also perform extremely well, with high AUC and ACC values. This proves that these methods are well-designed for detecting fake face images. The results obtained from experiments with the CelebDFv2 are often lower than the CelebA-HQ. This is believed to be due to the loss of significant contextual information when images are extracted from the original videos, making this dataset more challenging.

TABLE 4 In-Dataset Evaluation on the CelebA-HQ Subset of the ForensicsImage Dataset

Generative Method Method	LDM		DDIM		LDM-t2i	
	AUC	ACC	AUC	ACC	AUC	ACC
EfficientNetBo [60]	90.1	89.2	92.3	90.5	87.3	83.1
GramNet [37]	87.1	83.8	89.3	85.3	85.6	83.1
Fusing [36]	94.7	93.3	93.4	92.5	93.9	91.2
DIRE [18]	99.1	98.8	100.0	100.0	96.8	95.4
CNN-DCT [38]	97.5	96.6	97.7	96.8	95.3	93.5
DiffCoR (Ours)	100.0	100.0	99.8	99.7	99.7	99.3

Generative Method	Method	AUC	ACC
LDM	EfficientNetB0 [60]	87.9	83.4
	GramNet [37]	98.1	96.7
	Fusing [36]	97.3	95.7
	DIRE [18]	99.2	97.9
	CNN-DCT [38]	93.6	90.7
	DiffCoR (Ours)	100.0	100.0
Comparison of our DiffCoR model and other image detectors.			

TABLE 5 In-Dataset Evaluation on the CelebDFv2 Subset of the ForensicsImage Dataset

Method	AUC	ACC
EfficientNetB0 [60]	80.1	77.2
GramNet [37]	84.2	80.9
Fusing [36]	87.3	81.9
DIRE [18]	82.3	78.1
CNN-DCT [38]	85.7	79.2
DiffCoR (Ours)	88.9	81.2
Comparison of our DiffCoR model and other image detectors.		

To evaluate the comprehensive ability and scalability of the model when trained with diverse data types, we conduct experiments using the AI-Gen. Results in Table 6 show the performance of various detectors. The DiffCoR with EfficientNetBo backbone [60] demonstrates its robustness and superior performance with the highest AUC score of 92.1, indicating that it has the best overall ability to discriminate between original and manipulated images in this dataset. The DiffCoR only uses the ViT-B16 backbone, which also achieves similarly good results. However, this model requires more training time and takes longer to converge compared to the EfficientNetBo backbone. The DIRE [18] shows the highest accuracy, making it a strong competitor, particularly for tasks where ACC is prioritized. The results of the DIRE [18] fluctuate significantly, as this method requires selecting appropriate inversion and reconstruction models. This limitation becomes evident when experimenting with diverse data types in the AI-Gen.

TABLE 6 In-Dataset Evaluation on the AI-Gen Subset of the ForensicsImage Dataset

Generative Method	Method	AUC	ACC
Includes all generative methods in the AI-Gen	EfficientNetB0 [60]	87.3	80.1
	GramNet [37]	73.1	71.7
	Fusing [36]	87.1	80.9
	DIRE [18]	88.4	84.7
	CNN-DCT [38]	87.7	81.7
	DiffCoR (EfficientNetB0)	92.1	83.2
	DiffCoR (ViT-B16)	91.9	83.9
Comparison of our DiffCoR model and other image detectors.			

C. Cross-Dataset Evaluation

In this section, we run experiments across the dataset. Utilizing images from unknown sources in testing may lead to a reduction in the performance of the detection model. Therefore, we evaluate the effectiveness of the DiffCoR in cross-datasets to expose the generalization ability of these methods. In Tables 7 and 8, we present experimental results conducted by training on the LSUN-Bedroom LDM and testing on the LSUN-Bedroom DDPM, IDDPM, DDIM, GD, and LDM-t2i. Table 7 shows that the DiffCoR method achieves the highest AUC and accuracy scores across all three generative methods. Specifically, the DiffCoR achieves an AUC of 99.9, 99.4, and 97.4 for DDPM, IDDPM, and DDIM, respectively. the DIRE [18] method achieves the highest accuracy scores in the IDDPM and the DDIM. Table 8 shows that the DiffCoR method achieves the highest AUC across both GD and LDM-t2t for cross-testing. The DIRE [18] and Fusing [37] methods achieve similarly high results in cross-tests with all generative methods. However, the Fusing [37] method encounters some limitations during experiments, such as instability in training results. Additionally, for the DIRE [18] method to achieve high performance, it is essential to select appropriate inversion and generative methods. The EfficientNetBo [60], GramNet [37], and CNN-DCT [38] methods yield lower results. Notably, the performance of the CNN-DCT [38] method decreases significantly when converting and classifying in the frequency domain across different datasets.

TABLE 7 Cross-Dataset Evaluation on the LSUN-Bedroom Subset of the ForensicsImage Dataset

Generative Method	DDPM		IDDPM		DDIM	
Method	AUC	ACC	AUC	ACC	AUC	ACC
EfficientNetB0 [60]	81.1	73.3	84.7	76.6	83.4	74.5
GramNet [37]	88.1	81.8	89.3	80.3	87.4	82.3
Fusing [36]	90.8	89.7	91.4	88.1	90.3	80.4
DIRE [18]	92.1	85.8	98.2	94.7	94.3	85.5
CNN-DCT [38]	82.1	71.9	83.4	74.5	82.3	75.1
DiffCoR (Ours)	99.9	98.8	99.4	93.7	97.4	80.7

Using the training set of the LDM for training and the DDPM, the IDDPM, and the DDIM for cross-testing.

TABLE 8 Cross-Dataset Evaluation on the LSUN-Bedroom Subset of the ForensicsImage Dataset

Generative Method	GD		LDM-t2i	
Method	AUC	ACC	AUC	ACC
EfficientNetB0 [60]	81.7	72.8	83.7	71.9
GramNet [37]	85.3	77.8	85.6	81.3
Fusing [36]	86.4	79.3	84.4	80.1
DIRE [18]	93.2	78.8	91.1	84.4
CNN-DCT [38]	83.3	77.4	81.1	72.8
DiffCoR (Ours)	97.4	80.7	94.9	74.3

Using the training set of the LDM for training and the GD and the LDM-t2t for cross-testing.

As shown in Table 9, the evaluation was conducted using the training set of the CelebA-HQ LDM for training and DALL-E [3], Imagen [2], and Midjourney [52] for cross-testing. The DiffCoR attains the highest performance in DALL-E [3], and Midjourney [52]. The DIRE [18] achieves the best results in Imagen [2]. Table 10 presents the results of a cross-dataset evaluation, with training on AI-Gen and testing on LSUN-Bedroom DDPM and LSUN-Bedroom GD. The DiffCoR achieves the highest AUC and ACC scores across both the DDPM and the GD. Fusing [37] achieves the second-highest performance on both tasks. EfficientNetBo [60] and CNN-DCT [38] show lower performance on both tasks compared to the other models. These experiments demonstrate that DiffCoR improves the generalization of the model across a wide range of diverse datasets.

TABLE 9 Cross-Dataset Evaluation on the CelebA-HQ Subset of the ForensicsImage Dataset

Generative Method	DALL-E		Imagen		Midjourney	
Method	AUC	ACC	AUC	ACC	AUC	ACC
EfficientNetB0 [60]	67.5	61.8	66.8	60.8	61.8	59.3
GramNet [37]	77.3	69.8	74.2	71.7	79.3	73.4
Fusing [36]	82.4	72.9	74.9	71.2	74.9	69.2
DIRE [18]	82.1	74.4	84.1	75.4	80.2	74.9
CNN-DCT [38]	70.1	67.8	76.2	69.8	72.9	68.7
DiffCoR (Ours)	88.0	78.3	81.3	72.6	83.9	77.7

Using the training set of the LDM for training and the DALL-E, the imagen, and the Midjourney for cross-testing.

TABLE 10 Cross -Dataset Evaluation, Training on the AI-Gen Cross-Testing With the LSUN-Bedroom DDPM, and the LSUN-Bedroom GD

Generative Method	LSUN DDPM		LSUN GD	
Method	AUC	ACC	AUC	ACC
EfficientNetB0 [60]	71.3	66.4	77.8	65.5
GramNet [37]	79.7	71.4	76.3	68.3
Fusing [36]	82.2	80.3	84.7	81.2
DIRE [18]	84.3	81.8	83.7	80.7
CNN-DCT [38]	76.8	64.7	75.5	67.6
DiffCoR (Ours)	88.7	86.3	85.6	83.3

The strong cross-dataset performance of DiffCoR highlights its ability to generalize well across a variety of unseen generative models and datasets. This robustness is essential for real-world deployment, where the

origin of synthetic content is typically unknown and constantly evolving. Practical applications of DiffCoR include its integration into social media platforms for detecting manipulated images in user-generated content, news, and journalism verification pipelines to ensure the authenticity of visual media, and digital forensics investigations where reliable identification of tampered images is critical. Moreover, the modular nature of DiffCoR allows it to be fine-tuned or adapted to specific domains, making it a versatile solution for real-time and large-scale AI-generated image detection in the wild.

D. Ablation Study

Ablation study on data augmentation: We evaluate multiple data augmentation strategies: RRCAug, REAug, and our proposed RanSelect, using DiffCoR with $\lambda = 1$ and EfficientNetBo [60] on the AI-Gen dataset. As shown in Table 11, all augmentation methods improved AUC and ACC compared to training without augmentation (NoAug), confirming their effectiveness. DiffCoR consistently outperforms the baseline across all augmentations, with RanSelect achieving the best performance (AUC: 92.1, ACC: 83.2), demonstrating its advantage for this task.

TABLE 11 Ablation Study on Data Augmentation

Data Aug	Method	AUC	ACC
NoAug	Baseline	87.3	81.1
	DiffCoR	90.3	82.1
RRCAug	Baseline	88.6	81.8
	DiffCoR	90.3	82.1
REAug	Baseline	88.9	82.0
	DiffCoR	91.7	84.8
RanSelect	Baseline	90.3	82.8
	DiffCoR	92.1	83.2

Ablation study on the parameter λ : The results of the ablation study on the parameter λ are shown in Fig. 4. We investigate the effectiveness of the LCL on the proposal. We conduct experiments using the AI-Gen dataset, employing RanSelect as a data augmentation technique. In all cases, the model achieved the highest performance when λ was set to 1. This suggests that this value of λ may be the optimal setting used in this study. All remaining cases improve the quality of the model compared to when LCL is not applied, for instance, where the parameter λ is set to 10. When $\lambda = 0$, removing LCL leads to the lowest AUC and accuracy, indicating that classification alone is insufficient for robust generalization, overly large values (e.g., $\lambda = 5$ or $\lambda = 10$) slightly reduce performance, suggesting that excessive emphasis on consistency may weaken class discrimination. These results confirm that LCL, when properly balanced, significantly enhances model robustness and generalization. As demonstrated in this experiment, LCL contributes significantly to improving detection performance by promoting stable and discriminative feature learning across a wide range of generative models.

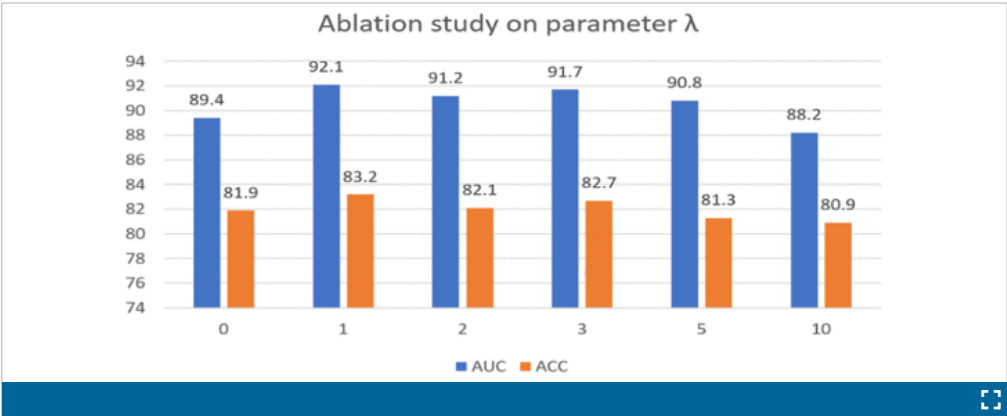


Figure 4. The results of the ablation study on the balance parameter λ .

E. Image Analysis

We do a frequency domain analysis of the data to obtain a better comprehension of the differences between the original and manipulated images. Fig. 5 shows the magnitudes of the Discrete Fourier Transform (DFT) for various images. We see that generated images show a much lower spectrum density going away from the center of the spectrum, which indicates dominating low-frequency components when compared to the original images. The spectrum distribution of generated images frequently exhibits greater disorder compared to the spectrum of the original image.

Fig. 5.

Some samples of the frequency analysis of the DiffCoR. We use the LSUN-Bedroom real images [50] and generate new images from the original LSUN-Bedroom dataset using generative techniques, DDPM [5], GD [6], DDIM [4], IDDPM [7], and LDM [1], LDM-t2i [11]. The x_0 , I_0 , and $\mathcal{F}$ represent the input image of the SDP, the input image of the IRL, and the spectrum of the image I_0 , respectively.

The average spectral differences across the three channels are illustrated in Fig. 6. In this figure, we utilized 1000 samples for each category: real, DDPM, GD, DDIM, LDM, and LDM-t2i. The spectral distribution results between the original images and the generated images at the input of SDP are distributed closely, which complicates the classification process. Conversely, at the input of IRL, the resulting spectral distribution shows greater separation, indicating an enhanced capability for classification.

Fig. 6.

The visualization of spectrum differences between original and generated images. "Real" represents the original images, and "fake" represents generated images from generative models. DDPM [5], GD [6], DDIM [4], IDDPM [7], LDM [1], and LDM-t2i [11]. We use 1000 samples for each technique. The x_0 and I_0 represent the input image of the SDP and the input image of the IRL, respectively.

Table 12 presents the results of a cross-dataset evaluation for different image classification models, focusing on distinguishing real from generated images. The models were trained on the Stable Diffusion v1.4 dataset and cross-tested on various other image generation models. DiffCoR outperformed other methods in terms of overall accuracy with an average accuracy (Avg ACC) of 77.5%, performing especially well across different subsets, including Stable Diffusion models and VQDM. UniversalFakeDetect and Swin-T also performed well, with Avg ACC values of 75.9% and 74.8%, respectively. Several methods, such as CNN Spot, F3Net, and Spec, showed lower average accuracy scores (around 64% to 69%), particularly struggling with datasets like GLIDE and BigGAN.

TABLE 12 Results of Cross -Dataset Evaluation on the GenImage Dataset [53], Training on SD v1.4, (Stable Diffusion v1.4) and Cross-Testing on Other Subsets

Method	Testing Subset								Avg ACC (%)
	Midjourney	SD v1.4	SD v1.5	ADM	GLIDE	Wukong	VQDM	BigGAN	
CNN Spot [32]	52.8	96.3	95.9	50.1	39.8	78.6	53.4	46.8	64.2
F3Net [16]	50.1	99.9	99.9	49.9	50.0	99.9	49.9	49.9	68.7
Spec [61]	52.0	99.4	99.2	49.7	49.8	94.8	55.6	49.8	68.8
GramNet [37]	54.2	99.2	99.1	50.3	54.6	98.9	50.8	51.7	69.9
DetT-S [62]	55.6	99.9	99.8	49.8	58.1	98.9	56.9	53.5	71.6
ResNet-50 [63]	54.9	99.9	99.7	53.5	61.9	98.2	56.6	52.0	72.1
Swin-T [64]	62.1	99.9	99.8	49.8	67.6	99.1	62.3	57.6	74.8
CLIPing the Deception [65]	59.6	99.7	98.9	53.4	64.4	96.9	63.7	58.9	74.4
UniversalFakeDetect [66]	61.2	98.8	97.9	57.7	68.9	97.8	61.7	63.3	75.9
DiffCoR (Ours)	66.4	99.9	98.9	61.8	69.6	97.2	65.0	61.2	77.5

To quantitatively assess the contribution of frequency domain analysis, we performed an ablation study by removing the Discrete Fourier Transform (DFT) components from the SDP module. As shown in Table 13, the exclusion of DFT-based features resulted in a drop in AUC from 92.1% to 89.9% and accuracy from 83.2% to 81.5%, confirming that frequency-based representations significantly enhance the model's ability to detect AI-generated images.

TABLE 13 Ablation Study on the Contribution of Frequency Domain Features (DFT)

Method	AUC	Accuracy
Full Model (w/ DFT)	92.1%	83.2%
Ablation (w/o DFT)	89.9%	81.5%

SECTION V. Conclusion

In this article, we presented DiffCoR, a novel detection method for identifying AI-generated images. Our approach leverages the unique properties of diffusion models and employs representation learning to maintain latent image consistency. We introduced the Latent Consistency Loss (LCL) to ensure that similar inputs have closely aligned latent representations, thereby enhancing structural and coherence properties in

the learned representations. Additionally, we conducted a comprehensive analysis of real and AI-generated images in the frequency domain using the Discrete Fourier Transform (DFT). This method provided deeper insights into the characteristics of AI-generated imagery and improved the detection accuracy for manipulated images. We also developed a new dataset, ForensicsImage, to serve as a benchmark for AI-generated image detectors. Our extensive experiments on the ForensicsImage dataset demonstrated that DiffCoR outperforms many recent approaches, achieving state-of-the-art performance in detecting AI-generated images. These contributions highlight the potential of DiffCoR as a robust tool for forensic analysis in the era of advanced generative models.

Currently, the DiffCoR primarily focuses on detecting AI-generated images. In the future, we plan to improve the method targeting AI-generated videos and audio, as well as evaluate these methods on more relevant datasets. Given the rapid advancements in generative AI, which are leading to increasingly sophisticated and high-quality samples, we will also develop corresponding methods to enhance detection capabilities for these types of content.

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